

RESPONSE TO REVIEWERS' COMMENTS

**Title Submission: AUTOMATIC SEGMENTATION OF MASSES ON
MAMMOGRAMS USING FUZZY LOGIC
ID Submission: 1721**

REVIEWER #1	AUTHOR'S RESPONSES
1. On what expert knowledge are the proposed rules (Rule 1, Rule 2, Rule 3) for Fuzzy Logic based?	In this paper, we propose fuzzy rules Rule1, Rule2, Rule3 based on the opinions of doctors and documents related to mammography at the address: https://radiologyassistant.nl/breast/bi-rads/bi-rads-for-mammography-and-ultrasound-2013 Based on the characteristics of the tumor, we proceed to build rules to retain important features in the preprocessing process and help identify the tumor better.
2. Are the thresholds like 200, $a=0.8$, $b=4$, etc., determined through experimental optimization or based on expert knowledge?	The thresholds given in the paper were tested many times after image preprocessing and segmentation results on the two proposed models, Mask R-CNN and U-net. The results demonstrate that the above threshold values give good segmentation results on the experimental dataset.
3. There is ample space in the paper for further discussion. The authors should provide more detailed explanations of their proposals and offer a discussion of the obtained results.	We have updated to the article content.
4. In section 2.5 (Evaluation Metrics), the authors mention three metrics: Accuracy, Dice coefficient, and DiceLoss. However, the experimental results only discuss Accuracy. Why are the other two metrics not addressed?	Dice Coefficient and Dice Loss are two important concepts in image segmentation, especially in applications such as medicine. In this paper, we also use this coefficient to evaluate the accuracy of the model shown in Figure 9.
5. The abstract and introduction suggest improved accuracy in locating masses, yet the experiments do not explicitly compare this aspect. How do the authors substantiate the claim of	We experimented on the same image dataset in 4 different scenarios: - 2 scenarios without using fuzzy logic in image preprocessing. - 2 scenarios using fuzzy logic in image preprocessing.

improved accuracy? Is this determined quantitatively or qualitatively?	The experimental results compared on 2 popular network models in medical image segmentation, U-net and Mask R-CNN. The experimental results show that when using fuzzy logic, the accuracy in image segmentation is significantly improved compared to the unused scenario.
6. Why did the authors choose these two methods for segmentation instead of other algorithms like Yolov10, RT-DETR, SAM, etc.?	After referring to the related of previous authors, the two network models U-net and Mask R-CNN are widely used and give quite good results on medical images, so we use these two network models.
7. Segmentation and detection tasks typically involve annotation. However, the paper does not clearly explain the annotation process or the tools used for it.	In this paper, we focus on identifying tumors on mammograms along with annotations on the accuracy between the detected image and the original image mask to compare the results.
8. The reviewer is unclear whether the focus of the paper is on segmentation or detection, as section 3.3 discusses both detection and classification.	Both detection and segmentation phases are very important in the process of detecting abnormalities in medical images. In the two network models we use, after training, the tumor on the input image is segmented.
REVIEWER #2	AUTHOR'S RESPONSES
- The intuition of choosing parameters in step 1, "Preprocessing and enhanced filtering."	- The thresholds given in the paper were tested many times after image preprocessing and segmentation results on the two proposed models, Mask R-CNN and U-net. The results demonstrate that the above threshold values give good segmentation results on the experimental dataset and are effective when compared with the scenario without using filters.
- In step 2, Fuzzy block, how do you determine fuzzy rules?	In this paper, we propose fuzzy rules Rule1, Rule2, Rule3 based on the opinions of doctors and documents related to mammography at the address: https://radiologyassistant.nl/breast/bi-rads/bi-rads-for-mammography-and-ultrasound-2013

	Based on the characteristics of the tumor, we proceed to build rules to retain important features in the preprocessing process and help identify the tumor better.
- At the Fuzzy block in Figure 7, please making a link between the steps and the below descriptions.	In Figure 7, we present in more detail the fuzzy block application phase shown in Figure 5. In this phase, we apply fuzzy rules -> the output image will be fed into 2 network models, U-net and Mask R-CNN, to extract features and train.
- Sections 3.2 and 3.3 have to be explained in more detail. In the current manuscript, I cannot understand what will happen at these phases.	Due to the limited number of pages of the article, we try to present the main contents of our research results. In section 3.2, after declaring the necessary training parameters described in Table 1, the data after preprocessing will be trained on 2 network models U-net and Mask R-CNN. To ensure fairness between the scenarios, we agree to use the same training parameters on 4 scenarios including: learning-rate=0.0001, epochs=100, mAP@IoU=0.5, num classes = 2. In section 3.3, after the training process on the image dataset, we will record the results including loss, accuracy, and trained models in file format.
- Please explain the obtained results from Figures 8 and 9.	- Refer to Fig. 8 for a chart displaying the model training over four scenarios. These training sets include the mask of the mass. The model effectively extracts features and learns the mass's characteristics, enhancing the learning capacity of each model through fuzzy logic techniques applied in image preprocessing. The models extract general features based on the mass's characteristics in the image. Training will stop when the Loss value no longer decreases, at which point we save the best model. The average TotalLoss values measured at this point are 0.4015, 0.3150, 0.1503, 0.1013 respectively. - In Figure 9, the experimental accuracy on the test set is presented. The results show that scenarios 3 and 4 after applying fuzzy logic give better results than when not applied on image data during preprocessing.

- Please improve the quality of Figures 8 and 9. For example, make the X-axis title and the numbers in both axes bigger.	We will edit and improve the image quality in these 2 figures.
- What is the title of the X-axis in Figure 10a) and the Y-axis in Figure 10b)?	In figure 10a) the X-axis represents the accuracy of the 4 experimental scenarios, in figure 10b) the Y-axis represents the training time of the 4 scenarios (hours).
- The title of section 2.1 should be Convolutional Neural Network Model. The references should be updated latest research in such field (e.g. refer to “A Research for Segmentation of Brain Tumors Based on GAN Model”, 2022)	In section 2.1 we want to present an overview of the techniques that will be used in the paper. We will also review recent studies to be able to evaluate the effectiveness of the proposed method.
REVIEWER #3	AUTHOR’S RESPONSES
In this paper, the authors introduce an automated breast mass detection system. The system integrates fuzzy logic for preprocessing images alongside deep learning models such as U-Net and Mask R-CNN for effective mass segmentation. The experimental findings demonstrate that this approach significantly enhances the quality of input images, preserves key features efficiently, and supports accurate mass localization. The system achieved an impressive accuracy of 92.5% on a dataset collected from various hospitals. These results provide a crucial basis for further refining fuzzy learning techniques aimed at the classification and identification of masses in mammograms.	Thank you for your encouragement. In this paper, we applied fuzzy logic and through experimental results proved its effectiveness on the same data set. In the future, we will continue to conduct further research by using more fuzzy learning techniques in the next stages including: feature extraction, training and classification to propose effective solutions in supporting doctors in diagnosing abnormalities on mammograms.